

IMPLEMENTATION HANDOFF

Frontend Blockers Resolution

Backend changes, API contracts, seeded test data, and verification record

Repository	teslimsadiqnevo/nevo-backend-2.0
Branch	main
Release commit	714861e - feat: resolve frontend integration blockers
Migration	20260828_0028_frontend_unblockers (applied)
Prepared	29 August 2026
Primary audience	Frontend engineering, QA, and deployment owners

Outcome

All backend items identified in blockers.pdf now have an available route, data contract, persistence layer, safe fallback, or seeded test fixture. The live read and write paths were exercised against Supabase before handoff.

1. Resolution Summary

01	Class student roster RESOLVED The class detail screen can now load its student roster from the backend. •Added GET /api/v1/classes/{class_id}/students. •Returns identity, account status, profile status, and latest lesson session time. •Checks the teacher's school context before returning class data.
02	Concept names in mastery responses RESOLVED Mastery responses now include human-readable concept names instead of UUID-only output. •Added a concepts reference table and concept lookup endpoints. •Student, class, and school mastery aggregation joins concept metadata. •A deterministic fallback label remains available for legacy rows without metadata.

Ask Nevo fallback returning the rendered prompt**RESOLVED****03**

A failed AI provider call can no longer echo prompt variables or internal context to the user.

- Removed user_content from the fallback response source.
- Ask Nevo now returns a neutral, safe retry message when the provider is unavailable.
- Added a regression test that rejects prompt headings, context JSON, and internal IDs.

Server-side lesson file upload**RESOLVED****04**

Teachers can upload source files directly to the parsing pipeline.

- Added POST /api/content/upload as multipart/form-data.
- Supports TXT, Markdown, PDF, DOCX, and PPTX extraction.
- Extracted content enters the existing content parsing service with source metadata.

Current teacher identity and profile**RESOLVED****05**

The frontend can retrieve the authenticated user's stable identity and school context.

- Added GET /api/v1/users/me.
- Returns user ID, role, names, display name, email, school, and subject list.
- Uses the same Bearer session as the existing authentication endpoints.

Lesson and block parsing access**RESOLVED****06**

Parsed lesson structures are now retrievable as lesson summaries and segment-level details.

- Added lesson listing and detail aliases under /api/content and /api/v1.
- Lesson detail includes ordered segments, modalities, comprehension checkpoints, and review reasons.
- Large source files still pass through the existing parser's lesson/module/segment boundary pipeline.

Demo teacher had no classes or usable console data**RESOLVED****07**

An idempotent seed now creates a complete shared-school console scenario.

- Teacher and admin are linked to Nevo Demo School and JSS 2A.
- Three students, enrollments, lesson content, concepts, mastery, scheduler, scaffold, and session data are seeded.
- Messages, notifications, and lesson assignments are included for realistic screen testing.
- Credentials can be reset through environment variables and are never committed to source.

08

Missing frontend service domains**RESOLVED**

Backend contracts now exist for the remaining screens identified by frontend engineering.

- Connect messaging and thread history.
- Notifications, unread count, and mark-as-read.
- Lesson library, lesson detail, upload, and assignment submission.
- Baseline/profile/settings persistence and password-reset request.
- Compatibility routes for analytics and intelligence screens.

2. API Contract Inventory

All routes below are available in Swagger. Unless noted otherwise, protected routes require Authorization: Bearer <access_token>. JSON responses use the aliases shown by Swagger.

Method	Endpoint	Frontend use
GET	/api/v1/users/me	Authenticated user identity, role, and school context
GET	/api/v1/classes/{class_id}/students	Roster for teacher class detail
GET	/api/concepts	Concept reference list
GET	/api/concepts/{concept_id}	Single concept metadata
GET	/api/content/lessons	Parsed lesson library
GET	/api/v1/lessons	Versioned lesson library alias
GET	/api/content/lessons/{lesson_id}	Lesson and ordered segment detail
GET	/api/v1/lessons/{lesson_id}	Versioned lesson detail alias
POST	/api/content/upload	Multipart source-file parsing
POST	/api/v1/lesson-assignments	Assign lesson to class or students
GET	/api/notifications	Notification feed and unread count
GET	/api/notifications/unread-count	Compact badge count
POST	/api/notifications/{notification_id}/read	Mark one notification as read
GET	/api/messages/threads	Connect conversation list

Method	Endpoint	Frontend use
GET	/api/messages/threads/{thread_id}	Chronological thread messages
POST	/api/messages	Create or continue student/class conversation
POST	/api/baseline/submit	Persist baseline feature summary
GET	/api/baseline/recalibrate-prompt/{student_id}	Next dimension prompt
GET	/api/intelligence/profile/{student_id}	Learner profile compatibility view
GET	/api/intelligence/flags	Flags filtered by student or class
GET	/api/intelligence/recommendations/{student_id}	Intervention recommendations
GET	/api/analytics/schools	School analytics readiness summary
GET	/api/analytics/outcomes	Outcome analytics compatibility response
GET	/api/settings/me	Current persisted user settings
PUT	/api/settings/me	Merge and persist user settings
POST	/api/v1/auth/forgot-password	Generic password-reset receipt
POST	/api/v1/auth/password-reset/request	Password-reset request alias

3. Important Payload Notes

Class roster

Each student row includes `student_id`, `first_name`, `last_name`, `display_name`, `login_identifier`, `status`, `profileStatus`, and `latestSessionAt`. The existing teacher-class assignment route retains its original `snake_case` response contract.

Mastery

Mastery state and aggregate responses now include `conceptName`. Frontend code should display that field and retain `conceptId` as the stable key.

Lesson upload

Send `multipart/form-data` with a single file field named `file`. The backend extracts readable text, records `filename/content type/byte length`, and submits it to the content parser. Unreadable or unsupported files return HTTP 400 with a functional error message.

Lesson assignment

Request fields are lessonId, optional classId, optional studentIds, and optional dueAt. When classId is supplied, current enrollments are expanded server-side. The response returns assignmentIds and createdCount.

Messaging

POST /api/messages accepts recipientId, recipientType (student or class), and content. The backend finds or creates the matching school-scoped thread, updates its preview, and returns the new message.

Password reset

The request route always returns a generic accepted receipt to avoid account enumeration. It creates a hashed, expiring reset token record for known eligible users. Email delivery and the final token-consumption screen remain deployment/integration responsibilities.

4. Persistence and Migration

Migration 0028 adds the persistence needed by these screens without replacing existing domain tables. It has already been applied to the configured Supabase database.

concepts	Human-readable concept metadata used by mastery and scheduler views
notifications	User-scoped notification feed and read state
message_threads	School-scoped Connect conversation metadata
messages	Chronological thread messages
lesson_assignments	Teacher-to-student/class lesson assignment records
password_reset_tokens	Hashed, expiring, single-use reset-token records
users.baseline_profile	Aggregated baseline/profile and settings JSONB state

Migration 0027 was also corrected to render through Alembic offline mode. This removes a MockConnection failure that previously caused every migration snapshot test to fail while leaving the live schema behavior unchanged.

5. Demo Data and Manual Test Setup

The idempotent seed script is `scripts/seed_demo_teacher_console.py`. It can be rerun safely and accepts optional environment variables for credential resets. No passwords, PINs, database URLs, peppers, or provider keys are stored in the repository or this handoff.

School	Nevo Demo School (school code NEVO-DEMO)
Class	JSS 2A
Teacher	Linked as primary teacher for JSS 2A
Students	Amara Okafor, Kofi Dada, and Dara Ibrahim
Lesson	Fractions Lesson 3 with parsed segments
Learning data	Concept mastery, FSRS due reviews, scaffold state/log, lesson sessions
Console data	Assignments, notification, Connect thread, and message

6. Verification Record

Static checks	Ruff, compileall, and git diff checks passed
Focused tests	14 auth, Ask Nevo, mastery, and parsing tests passed
Migration tests	47 migration contract tests passed
Effective suite	404 tests passing; 25 PostgreSQL integration tests skipped without TEST_DATABASE_URL
Live read paths	Teacher, admin, and student authenticated endpoint sweep returned HTTP 200
Live write paths	Upload 200, assignment 201, message 201, settings/baseline 200, notification read 204
Swagger	87 documented API paths; blocker routes tagged Frontend Unblockers
Database	Supabase migration head confirmed at 20260828_0028

7. Frontend Integration Guidance

- Use `NEXT_PUBLIC_API_URL` for the backend origin. Local development currently targets `http://localhost:8000`.
- Send the Bearer token on every protected request. The backend remains the authorization boundary.
- Use Swagger response aliases exactly. Some established routes remain snake_case while new compatibility routes expose camelCase aliases.

- Treat 404 as missing resource, 403 as wrong role/school context, and 503 as a temporarily unavailable domain service.
- Analytics compatibility routes are available for integration but currently return readiness/empty outcome structures until production outcome data is populated.
- The password-reset request is backend-ready; transactional email delivery and final reset-token consumption still need the selected email provider and frontend reset form.
- Credentials for shared test accounts should be exchanged privately, not committed or copied into tickets.

8. Deployment Notes

Build command: `python -m pip install --upgrade pip && python -m pip install .`

Start command: `alembic upgrade head && uvicorn nevo.main:app --host 0.0.0.0 --port $PORT`

The pushed release is commit 714861e on main. Render or another deployment platform should run the migration before starting the API. Required production secrets remain environment variables and are intentionally excluded from this document.

9. Acceptance Checklist

- Teacher can sign in and GET `/api/v1/users/me`.
- Teacher class list contains JSS 2A and its roster contains three students.
- Mastery UI displays conceptName instead of a raw UUID.
- Lesson library and lesson detail load from backend data.
- TXT/PDF/DOCX/PPTX upload reaches the parsing pipeline.
- Lesson assignment returns created assignment IDs.
- Notifications and Connect threads load and write successfully.
- Ask Nevo provider failure never reveals the rendered prompt or context JSON.
- Settings and baseline submissions persist across requests.
- Swagger shows the Frontend Unblockers route group.